Composite Nacelle Fairings – General data

1. General

The composite nacelle fairings are installed on bottom surface of the wing aft of the main engine nacelle structure.

2. Description

The composite nacelle fairings are attached to the aluminum airframe using perimeter 10 - 32 screws spaced approximately every 3 inches. The fairings are attached to the metal firewall on the forward edge and to the lower wing surface on the sides and aft edge. Expanded foil is applied from the forward to the aft end of the fairings so that it is captured by the mounting screws. The foil stops just short of the side flanges for manufacturing reasons. No conductive path to the airframe is provided along the side attachment fasteners. The forward and aft fastener locations are considered the principal lightning energy transfer paths. A solid aluminum strip is bonded into the aft edge of the fairing to assist in energy conduction.

3. Removal/Installation

Refer to Figure 1 for details.

A. Removal of Composite Nacelle Fairings

1) Remove the screws from the strut-to-nacelle.

2) Pull the strut-to-nacelle fairing away from the nacelle fairing.

3) Remove the screws from the fairing that attach the fairing to the wing.

4) Remove the screws from the nacelle fairing.

5) Remove the nacelle fairing from the main engine nacelle structure.

B. Installation of Composite Nacelle Fairings

1) Position the nacelle fairing with the main engine nacelle structure.

2) Install the screws on the nacelle fairing and the fairing.

Note For aircraft incorporating Mod 6/2340, the fairings have rubber washers installed, torque the screws to compress the rubber washers (which are underneath the metal washers) to approximately one-half of their original thickness. For fairings without rubber washers (Pre Mod 6/2340) torque the #10 screws to 10 – 16 in–lbs.

3) Position the strut-to-nacelle fairing on the nacelle fairing.

4) Install the screws on the strut-to-nacelle fairing.

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5) Do a bonding test (refer to Para 4.).

4. Adjustment/Test

Refer to Figure 2 for details.

A. Composite Nacelle Fairing Bonding Test

1) With a milliohm meter, measure the resistance between the nacelle fairing and the airframe. Resistance must be less than 10 milliohms.

Note Probe measurements to be taken on one of the strut-to-nacelle fairing screws or nut-plate fasteners and an airframe side blind rivet.

2) If the measurement is more than the bonding requirement of less than 10 milliohms, do the Bonding Adjustment/Rework (refer to Para 4.B).

B. Composite Nacelle Bonding Adjustment/Rework

1) Remove the nacelle fairing forward and aft attachment screws and clean the screws as required to achieve bonding requirement.

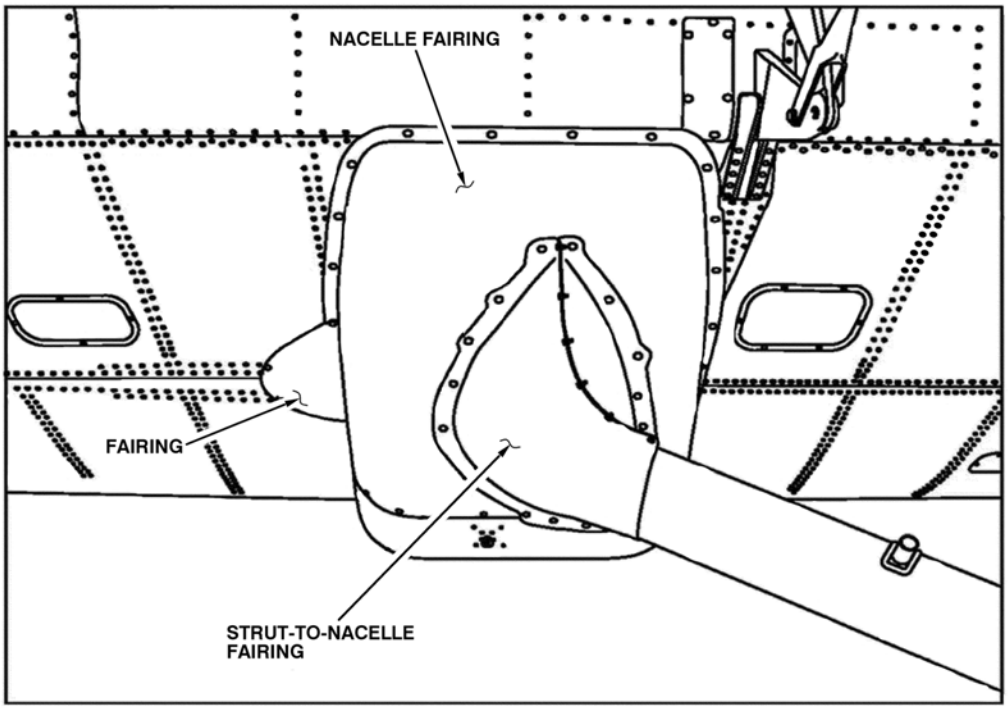
2) Remove the strut-to-nacelle fairing screws and clean the screws as required to achieve bonding requirement.

3) If required, remove the composite nacelle fairing (refer to Para 3.A).

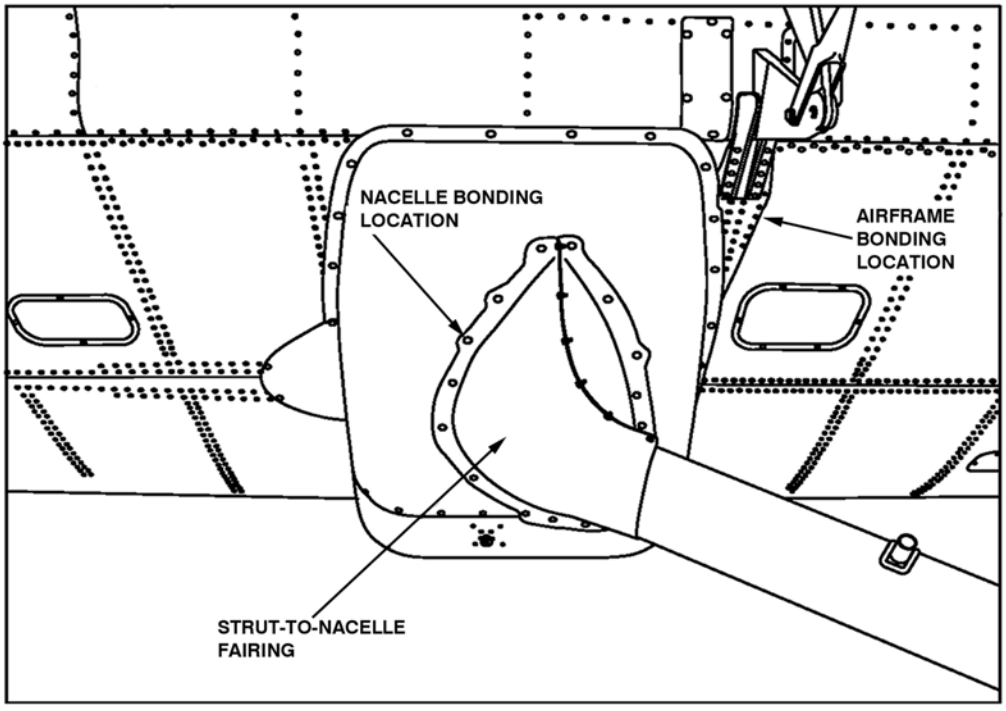
4) Clean the aft mating surface of the nacelle aluminum strip.

5) If removed, install the composite nacelle fairing (refer to Para 3.B).

Note For reference, see applicable standard electrical bonding procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data.



Composite Nacelle Fairing – Removal/Installation
Figure 1



Composite Nacelle Fairing – Bonding Locations
Figure 2

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